
MODULE *Blockchain*

EXTENDS *Integers, Sequences, TLC, Naturals*

CONSTANTS *Customers*

VARIABLES *ledger, contractStates, actions*

EventTypesStrings $\triangleq \{\text{"anomaly"}, \text{"pauseBilling"}\}$

States $\triangleq \{\text{"ready"}, \text{"actioned"}\}$

Init $\triangleq$

$\wedge \text{ledger} = \langle \rangle$

$\wedge \text{contractStates} = [c \in \text{Customers} \mapsto \text{"ready"}]$

$\wedge \text{actions} = \langle \rangle$

RecordAnomaly(customer, eventTypeStr) $\triangleq$

$\wedge \text{eventTypeStr} = \text{"anomaly"}$

$\wedge \text{ledger}' = \text{Append}(\text{ledger}, [\text{type} \mapsto \text{eventTypeStr}, \text{customerId} \mapsto \text{customer}])$

$\wedge \text{UNCHANGED } \langle \text{contractStates}, \text{actions} \rangle$

TriggerContract $\triangleq$

$\exists i \in 1 \dots \text{Len}(\text{ledger}) :$

$\wedge \text{ledger}[i].\text{type} = \text{"anomaly"}$

$\wedge \text{contractStates}' = [\text{contractStates} \text{ EXCEPT } ![\text{ledger}[i].\text{customerId}] = \text{"actioned"}]$

$\wedge \text{actions}' = \text{Append}(\text{actions}, [\text{type} \mapsto \text{"pauseBilling"}, \text{customerId} \mapsto \text{ledger}[i].\text{customerId}])$

$\wedge \text{ledger}' = \text{ledger}$

Next $\triangleq$

$(\exists c \in \text{Customers}, \text{etStr} \in \text{EventTypesStrings} : \text{RecordAnomaly}(c, \text{etStr}))$

$\vee \text{TriggerContract}$

vars $\triangleq \langle \text{ledger}, \text{contractStates}, \text{actions} \rangle$

Spec $\triangleq \text{Init} \wedge \Box[\text{Next}]_{\text{vars}}$

Properties

LedgerImmutability $\triangleq \forall i \in 1 \dots \text{Len}(\text{ledger}) :$

$\forall j \in i \dots \text{Len}(\text{ledger}) : \text{ledger}[i] = \text{ledger}[j]$

ContractStateInvariant $\triangleq \forall c \in \text{Customers} : \text{contractStates}[c] = \text{"actioned"}$

$\Rightarrow \exists i \in 1 \dots \text{Len}(\text{ledger}) :$

$\text{ledger}[i].\text{customerId} = c \wedge \text{ledger}[i].\text{type} = \text{"anomaly"}$

TypeOK $\triangleq$

$\wedge \text{ledger} \in \text{Seq}([\text{type} : \{\text{"ready"}, \text{"actioned"}\}, \text{customerId} : \text{Customers}])$

$\wedge \text{contractStates} \in [\text{Customers} \rightarrow \text{States}]$

$\wedge \text{actions} \in \text{Seq}([\text{type} : \{\text{"ready"}, \text{"actioned"}\}, \text{customerId} : \text{Customers}])$

INVARIANTS $\triangleq$

$\text{TypeOK} \wedge \text{LedgerImmutability} \wedge \text{ContractStateInvariant}$

(* Renamed to avoid clash, parameterized *)
(* Action is only enabled if *eventTypeStr* is “**anomaly**” and leads to a state change. *)
(* Otherwise, this specific action instance is disabled. *)